



Hidden Subgroup Analysis under Missing Metadata in Medical Imaging

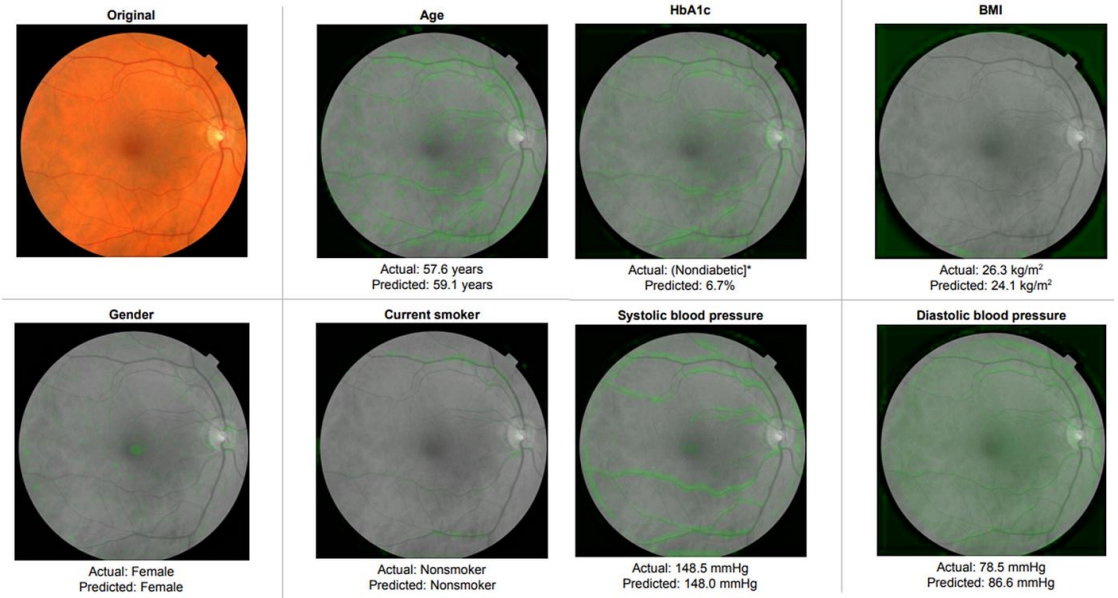
Yan Li(liyanly@bupt.edu.cn)

Yawen Li, **Yan Li**, Zhe Xue, Yingxia Shao, Meiyu Liang, Guanhua Ye. Beyond Metadata: CAPRA for Hidden Subgroup Analysis under Missing Metadata in Medical Imaging. ACM MM 2026.

More Than Disease

ACM Multimedia 2026
Rio de Janeiro, Brazil

Poplin et al. (Nature Biomedical Engineering, 2018) demonstrated that a single **retinal fundus image contains predictive information about multiple patient attributes**, such as age and sex.



Is Accuracy Enough?

ACM Multimedia 2026
Rio de Janeiro, Brazil



Is 70.8% enough to trust the model?

* DPE = Diverse Prototypical Ensembles (To et al., 2025).
Image-only setting; 7-Class classification.

Scores Hide Failures

ACM Multimedia 2026
Rio de Janeiro, Brazil



→ **DPE** →

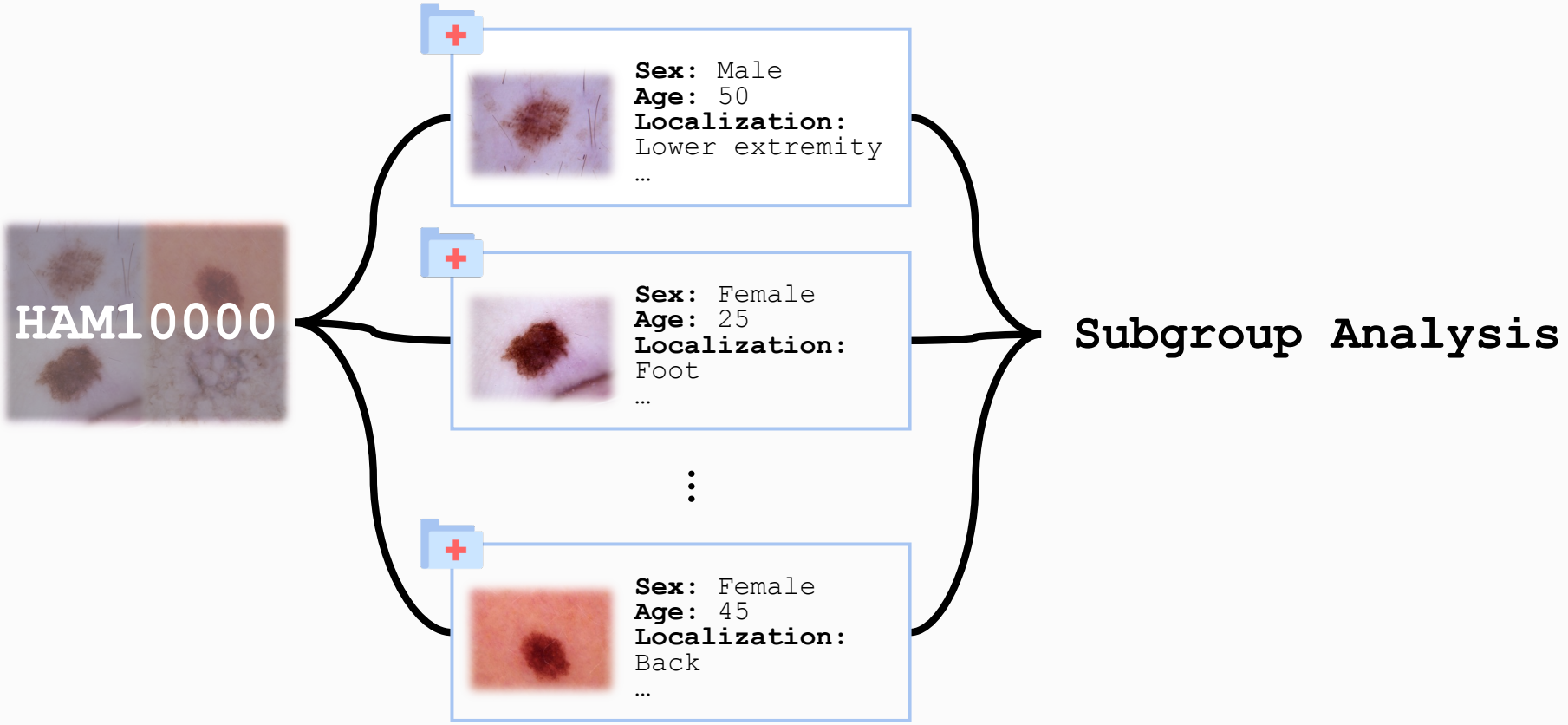
Lesion Location	BA	
Neck	79.3%	
Trunk	72.2%	
Back	71.4%	
	70.8%	**← Overall BA**
Face	68.0%	
Chest	65.2%	
Foot	**39.1%**	**← Worst subgroup**

-31.7 pts vs. overall

A model with reasonable aggregate performance can still fail severely on a specific subgroup.

Metadata-Driven Analysis

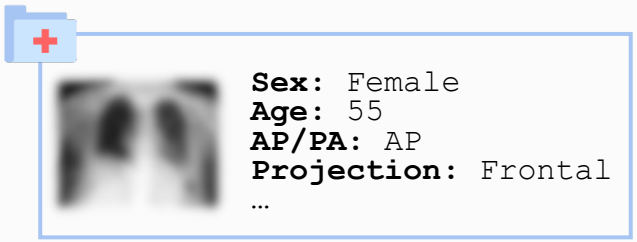
ACM Multimedia 2026
Rio de Janeiro, Brazil

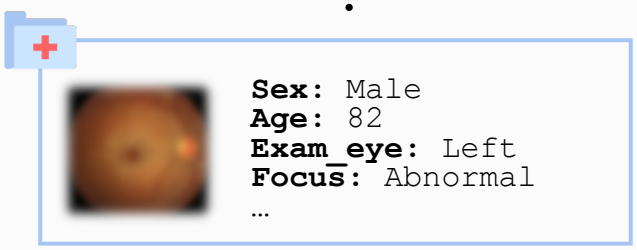


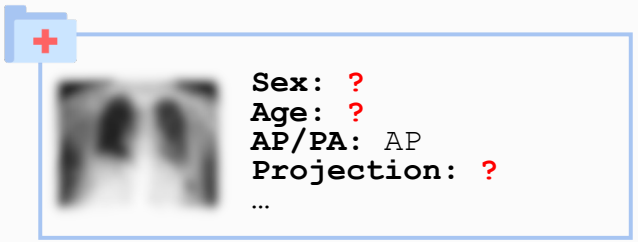
Deployment Gap

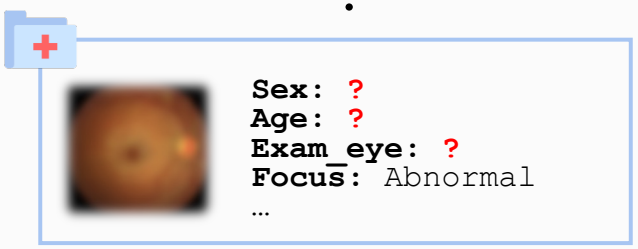
ACM Multimedia 2026
Rio de Janeiro, Brazil

Conventional Development Vs. Real-world Development









The image remains. The subgroup labels do not.

Subgroup Risk Analysis



Problem Formulation:

Given a medical image ***X***, task label ***Y***, conventional subgroup analysis relies on explicit metadata:

$$A = (A_1, \ldots, A_K), \qquad A_k \in V_k$$

A_k: The k-th subgroup attribute (e.g., age or sex).

The metadata-defined subgroup risk is:

$$\boxed{R^A_{k,v}(h) = \mathbb{E}\left[\ell(h(X), Y) \mid A_k = v\right]}$$

h: Task prediction model (e.g., CNN or Vision Transformer);

h(X): Predicted probability distribution over disease classes.

From Metadata- to …



Instead of relying on predefined subgroup attributes, latent-based methods discover groups from learned representations or model behavior.

$$z_i = \phi(X_i) \qquad \hat{G}_i = \mathrm{Cluster}(z_i)$$

The latent-based subgroup risk is:

$$\hat{R}_{G_j}(h) = \frac{1}{|G_j|} \sum_{i \in G_j} \ell(h(X_i), Y_i)$$

- No explicit subgroup metadata required for discovery.
- Discovered groups are not guaranteed to align with predefined clinical audit attributes.

From Metadata- to …



Proxy-Based Subgroup Analysis:

When explicit metadata A_k are unavailable, a straightforward approach is to infer subgroup attributes from medical images::

$$\hat{A}_k(X) = \arg\max_{v \in V_k} s_v^{(k)}(X)$$

The inferred attributes can then define proxy subgroups:

$$\widehat{G}_{k,v} = \{i : \hat{A}_k(X_i) = v\}$$

- Hard pseudo-labels discard subgroup uncertainty and may introduce misassignment errors.(SSA, ICLR 2022).
- Soft memberships capture uncertainty, but do not by themselves guarantee reliable subgroup risk estimates.(PG-DRO, AAAI 2023) .

Existing Methods

ACM Multimedia 2026
Rio de Janeiro, Brazil

Metadata-base ---------- **Metadata Dependency**

Explict Metadata $\boldsymbol{A}$ → Subgroup Anaysis

↓

Latent-base ---------- **Uncertain Semantic**

Image/Model Feature $\boldsymbol{X}$ → Discovered Group $\boldsymbol{G}$ → Subgroup Anaysis

↓

Proxy-base ---------- **Uncertain Proxy Reliability**

Image/Model Feature $\boldsymbol{X}$ → Predicted Attributes $\widehat{\boldsymbol{A}}$ → Subgroup Anaysis

↓

Which proxy axes are both reliable and failure-relevant?

CAPRA: Overview



Problem Setting:

We consider a task-labeled training set without subgroup annotations:

$$\mathcal{D}_{\mathrm{tr}} = \{(x_i, y_i)\}_{i=1}^{n}$$

and a small metadata-labeled calibration cohort:

$$\mathcal{D}_{\mathrm{cal}} = \{(x_j, y_j, a_j)\}_{j=1}^{m}, \qquad m \ll n.$$

At deployment, explicit subgroup metadata $\boldsymbol{A}$ are unavailable.

We therefore construct:

- ❑ $\boldsymbol{t}$: Semantic tokens from proxy predictions
- ❑ $\boldsymbol{q}$: Calibrated soft proxy posteriors
- ❑ $\boldsymbol{r}$: Sample-level posterior reliability
- ❑ $\boldsymbol{\alpha}$: Trust- and failure-aware axis weights

Calibrated Semantic Proxies



For each predefined subgroup axis k, we employ a proxy teacher T_k:

$$\boxed{s^{(k)}(X) = T_k(X)}$$

Posterior Calibration:

Using a small metadata-labeled calibration cohort, CAPRA combines temperature scaling with confusion-matrix shrinkage:

$$\boxed{q_{i,v}^{(k)} = (1-\lambda_k)\,\mathrm{softmax}\left(\frac{s_i^{(k)}}{\tau_k}\right)_v + \lambda_k C_{k,\hat{v}_i,v}}$$

Trust × Failure Relevance



Not all calibrated proxy axes are equally useful for subgroup analysis and robust learning.

We consider two complementary factors:

Axis Trust:

A proxy axis receives high trust only when its attribute predictions are both discriminative and reasonably calibrated.

$$u_k = \underbrace{\left[\frac{\mathrm{BA}_k - 1/K_k}{1 - 1/K_k}\right]_+}_{\text{Above-Chance Recognition}} \cdot \underbrace{[1 - \mathrm{ECE}_k]_+}_{\text{Calibration Quality}}$$

Trust × Failure Relevance



Not all calibrated proxy axes are equally useful for subgroup analysis and robust learning.

We consider two complementary factors:

Failure Relevance:

A higher $\boldsymbol{\Delta_k}$ indicates that the model's classification errors are more concentrated in at least one supported bucket along axis $\boldsymbol{k}$.

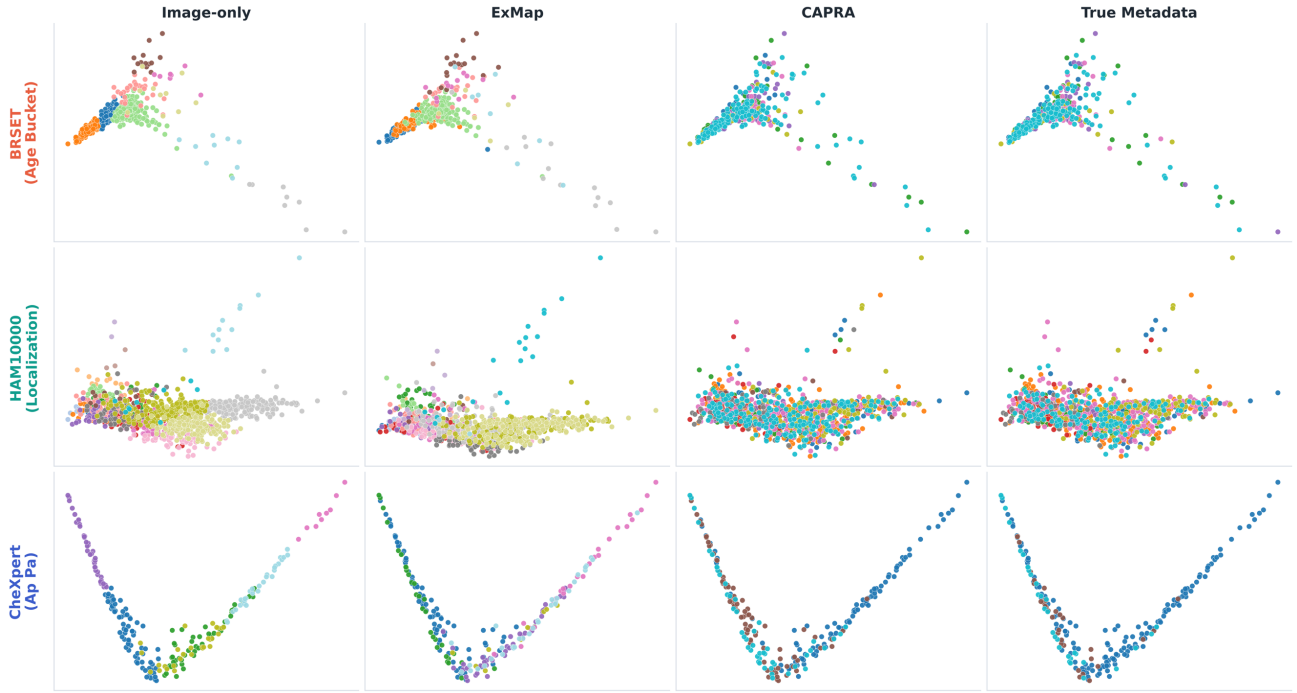
$$\Delta_k = \underbrace{\max_{v \in V_k^*} \frac{\sum_i q_{i,k,v}^{\mathrm{OOF}} \ell_i}{\sum_i q_{i,k,v}^{\mathrm{OOF}}}}_{\text{Worst Proxy-Bucket Error}} - \underbrace{\frac{\sum_{v \in V_k^*} \sum_i q_{i,k,v}^{\mathrm{OOF}} \ell_i}{\sum_{v \in V_k^*} \sum_i q_{i,k,v}^{\mathrm{OOF}}}}_{\text{Support-Weighted Mean Error}}$$

Semantic Alignment



CAPRA produces subgroup structures that are more closely aligned with predefined audit semantics than image-only and latent alternatives on NMI.

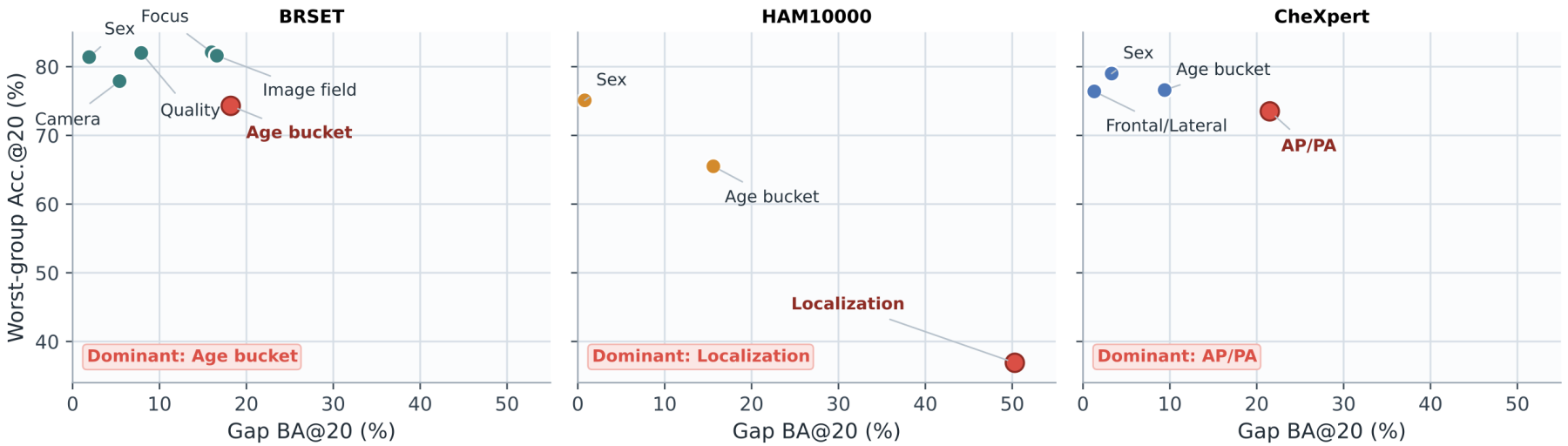
Source	BRSET			HAM10000			CheXpert		
	Pur./NMI ↑	Agr. (%) ↑	Gap (%) ↓	Pur./NMI ↑	Agr. (%) ↑	Gap (%) ↓	Pur./NMI ↑	Agr. (%) ↑	Gap (%) ↓
True metadata	**0.98 / 1.00**	**100.0**	**4.9**	**0.95 / 1.00**	**100.0**	**33.4**	**0.98 / 1.00**	**100.0**	**6.3**
Image-only	0.24 / 0.08	37.6	38.6	0.21 / 0.12	29.1	51.0	0.31 / 0.11	72.2	81.3
ExMap	0.08 / 0.06	37.4	36.3	0.23 / 0.11	27.8	51.4	**0.61** / 0.33	**84.2**	63.1
CAPRA	**0.81** / **0.60**	**83.3**	**19.0**	**0.68** / **0.68**	**81.7**	**10.8**	0.55 / **0.43**	70.5	**22.5**



Subgroup Failure Patterns

ACM Multimedia 2026
Rio de Janeiro, Brazil

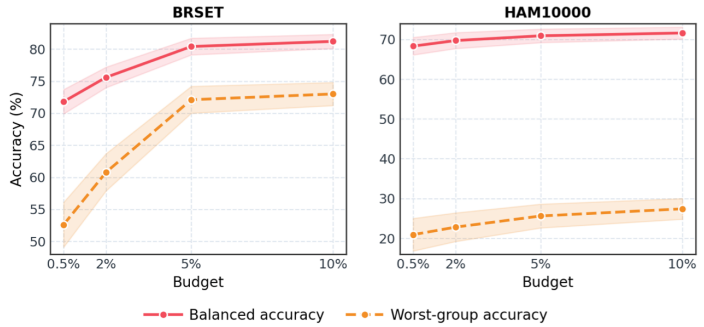
The dominant failure axis varies across modalities, highlighting the need for domain-adaptive, semantically interpretable subgroup analysis.



Points farther right and lower indicate more severe explicit subgroup failure.

Practicality

CAPRA achieves strong performance with limited metadata supervision.



Generalization

CAPRA demonstrates promising generalization under external imaging shifts.

Method	BRSET audit BA (%)	BRSET audit WGA (%)	mBRSET BA (%)	mBRSET Macro-F1 (%)
CAPRA anchor	**81.6 ± 1.6**	**72.9 ± 2.2**	66.8 ± 3.1	52.6 ± 3.7
CAPRA (ours)	80.2 ± 2.0	71.1 ± 2.5	**69.1 ± 2.7**	**60.8 ± 4.2**

Robust Adaptation



From Axis Importance to Sample Reweighting

CAPRA combines three complementary factors to construct sample-level weights:

$$\hat{w}_i = 1 + \lambda \sum_{k=1}^{K} \underbrace{\alpha_k}_{\text{Axis Importance}} \cdot \underbrace{r_i^{(k)}}_{\text{Sample Reliability}} \cdot \underbrace{e_i^{(k)}}_{\text{Excess Risk}}$$

Robust Adaptation Objective

After weight clipping and batch normalization, CAPRA optimizes:

$$\mathcal{L}_{\text{adapt}} = \frac{1}{|B_t|} \sum_{i \in B_t} w_i \, \ell(h(X_i), Y_i)$$

Downstream Gains



CAPRA provides reusable subgroup information that can improve worst-group robustness across different learners and imaging tasks.

Method	BRSET				HAM10000				CheXpert			
	BA (%)	Δ	WGA (%)	Δ	BA (%)	Δ	WGA (%)	Δ	BA (%)	Δ	WGA (%)	Δ
GroupDRO*	62.6±2.1	–	51.5±3.6	–	33.8±2.7	–	20.5±4.2	–	65.6±1.3	–	48.1±2.4	–
+ CAPRA	83.1±1.4	+20.5	69.9±2.0	+18.4	72.9±1.7	+39.1	33.2±3.3	+12.7	82.5±0.8	+16.9	61.4±1.9	+13.3
JTT	73.2±2.6	–	60.3±3.2	–	56.1±2.0	–	27.6±4.0	–	81.0±1.2	–	61.5±2.1	–
+ CAPRA	77.9±1.5	+4.7	73.2±2.3	+12.9	69.5±2.1	+13.4	41.0±2.7	+13.4	78.6±0.9	-2.4	66.9±1.7	+5.4
DFR	72.9±1.6	–	65.7±2.0	–	68.7±1.4	–	26.8±2.9	–	**89.5±0.7**	–	**77.7±1.3**	–
+ CAPRA	74.3±1.1	+1.4	70.4±1.9	+4.7	69.0±1.5	+0.3	28.0±2.1	+1.2	81.0±0.9	-8.5	74.5±1.2	-3.2
DPE	84.6±1.2	–	66.9±2.2	–	71.9±1.6	–	39.1±3.0	–	84.6±1.0	–	69.6±1.7	–
+ CAPRA	**86.0±0.7**	+1.4	**79.0±1.7**	+12.1	**74.5±1.1**	+2.6	**49.4±2.4**	+10.3	85.8±0.6	+1.2	71.5±1.8	+1.9
GSR	76.8±1.9	–	61.0±2.7	–	58.6±2.3	–	34.1±3.5	–	78.7±1.1	–	66.2±2.2	–
+ CAPRA	72.5±1.4	-4.3	64.4±2.6	+3.4	58.3±1.7	-0.3	41.1±2.9	+7.0	80.3±1.2	+1.6	68.3±1.6	+2.1

Future Directions



- **Generalization:** Adapt to diverse clinical environments and distribution shifts.
- **Trustworthiness:** Enable fine-grained and continuous model evaluation.
- **Clinical Deployment:** Integrate reliable AI into real-world clinical workflows.

Summary

- **Subgroup Analysis:** Enable meaningful subgroup analysis without deployment metadata.
- **Calibrated Proxy Axes:** Combine semantic information, uncertainty calibration, and trust-failure-aware weighting.
- **Robust Learning:** Demonstrate semantic alignment and improve worst-group robustness in most evaluated settings.

Thank You!





Contact: Yan Li(liyanly@bupt.edu.cn)

Yawen Li, **Yan Li**, Zhe Xue, Yingxia Shao, Meiyu Liang, Guanhua Ye. Beyond Metadata: CAPRA for Hidden Subgroup Analysis under Missing Metadata in Medical Imaging. ACM MM 2026.